

Numpy.

→ Numerical Python.

→ fundamental python library for numerical computations.

→ Mathematical functions.

→ Efficient multi-dimensional Array.

→ Linear Algebra

→ Broadcasting

→ Random no's generation

Numpy Array

(vs)

Python Lists

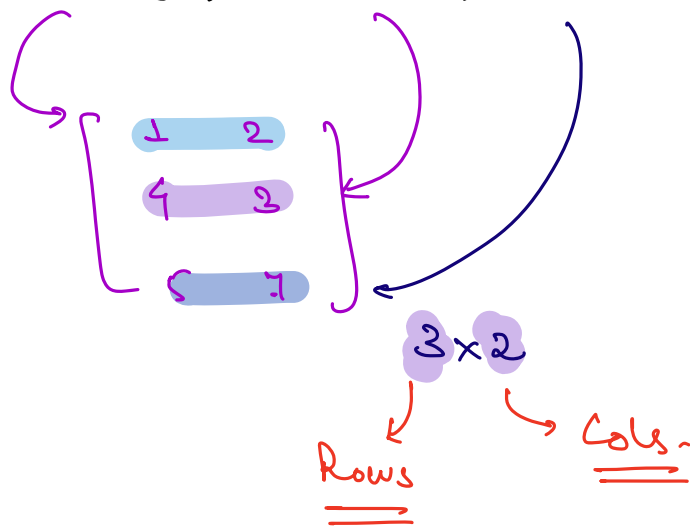


→ Slow.

→ Not memory efficient.

$[1 \ 2 \ 3] \Rightarrow 1D \text{ Array}$

$[[1, 2], [4, 3], [5, 7]]$



$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 8 & 4 & 1 \end{bmatrix} + \begin{Bmatrix} \end{Bmatrix} \times \text{error}$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} + \begin{bmatrix} 4 & 1 \\ 2 & 7 \\ 6 & 2 \end{bmatrix}_{3 \times 2} \Rightarrow \begin{bmatrix} 5 & 3 \\ 5 & 11 \\ 11 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} + \begin{bmatrix} 4 & 1 \\ 2 & 7 \\ 6 & 2 \\ 5 & 1 \end{bmatrix}_{4 \times 2} \Rightarrow \text{error}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} + [5 \ 6]_{1 \times 2} \Rightarrow \begin{bmatrix} 6 & 8 \\ 8 & 10 \\ 10 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 5 & 6 \\ 5 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} + \begin{bmatrix} 5 & 6 \\ 1 & 2 \end{bmatrix}_{2 \times 2} \rightarrow \text{Error}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} + \begin{bmatrix} 3 & 3 \\ 5 & 5 \\ 1 & 1 \end{bmatrix}_{3 \times 2}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} + \begin{bmatrix} 3 \\ 5 \\ 1 \\ 2 \end{bmatrix}$$